

MD. SHOUROB MAHMUD

Final-Year B.Sc. Student | Mechanical Engineering | RUET

Shourobmahmud.ruet@gmail.com | Rajshahi University of Engineering & Technology, Bangladesh

CAREER OBJECTIVE

Final-year Mechanical Engineering student with strong academic performance (CGPA 3.91/4.00) and hands-on experience in simulation, embedded systems, industrial training, and active research. Passionate about advanced bio-composite materials, intelligent automation, and sustainable engineering solutions. Seeking a high-impact engineering or research role to apply and expand expertise in design, analysis, and emerging technologies.

EDUCATION

B.Sc. in Mechanical Engineering 2021 – Present (Expected 2027)

Rajshahi University of Engineering & Technology (RUET), Bangladesh

- **Academic Performance:** CGPA: 3.91 / 4.00
- Core Areas: Structural Mechanics, Composite Materials, Thermodynamics, Fluid Mechanics, Control Systems, Robotics & Automation

RESEARCH EXPERIENCE

1. Undergraduate Thesis (Ongoing) 2026 – Present

Department of Mechanical Engineering, RUET

- **Bamboo–Jute Nano Clay Epoxy Resin Bio-Composite for Structural Applications**
 - Fabricating hybrid bio-composites with bamboo fiber, jute fiber, and nanoclay (and Al_2O_3) reinforcement in an epoxy matrix
 - Characterizing mechanical, thermal, and moisture-resistance properties via standardized testing protocols

2. Review Article (Ongoing — Under Preparation) 2025 – Present

Biomimetic Caudal Fin (BCF) Underwater Robot Locomotion

- Systematic review of fish-inspired BCF propulsion for autonomous underwater vehicles (AUVs)
 - Surveying 80+ publications on hydrodynamic modeling, thrust efficiency, and control strategies
 - Comparing soft-robotic and rigid-body fin designs across different operational environments

3. Research Paper (Ongoing) 2025 – Present

AI-Based Fault Detection in Renewable Energy Systems

- Developing ML-based fault detection framework for solar PV and wind turbine systems
 - Implementing SVM, Random Forest, and CNN models for real-time anomaly classification from sensor data
 - Benchmarking algorithm accuracy and generalizability across multiple renewable energy datasets

TECHNICAL SKILLS

Simulation & Analysis: ANSYS (Structural, Thermal, CFD basics), FEA, Mesh Convergence, Verification & Validation (V&V), Stress Analysis

Programming & IoT: Python (Basic), Arduino, ESP32, Proteus (Circuit Simulation)

Sensors & Modules: RFID, GSM, DHT11, MQ-2, Soil Moisture, LCD (I2C), OLED (SH1106), Relay Systems

Research Tools: Microsoft Excel (Advanced), Technical Report Writing, Literature Review, Data Analysis

Other: 3D Printing Technologies, Advanced Materials Characterization, Basic CAD

ENGINEERING PROJECTS

- ▶ **Smart Watering System — IoT-Based Automated Irrigation (ESP32)**
 - Designed automated irrigation system using ESP32 with soil moisture sensing and relay-controlled pump
 - Implemented real-time monitoring logic for autonomous water delivery based on soil condition thresholds
- ▶ **Multi-Access Security Control System (RFID + Fingerprint + Keypad)**
 - Developed multi-factor authentication system controlling a solenoid lock via RFID, keypad, and fingerprint modules
 - Simulated and validated complete circuit using Proteus prior to hardware deployment
 - Implemented secure access logic with fail-safe lock mechanism and unauthorized access alerting
- ▶ **ANSYS Structural Simulation Studies**
 - Performed structural analysis of beam and frame systems under various loading conditions
 - Conducted mesh convergence studies to ensure solution accuracy and reliability
 - Applied V&V methodology to validate simulation results against analytical benchmarks
- ▶ **ESP32 Environmental Monitoring Station**
 - Built multi-sensor node (temperature, humidity, air quality) with Wi-Fi cloud data transmission
 - Deployed real-time dashboard with threshold-based alerting for indoor environment monitoring

INDUSTRIAL TRAINING

Bangladesh Industrial Technical Assistance Center (BITAC), Dhaka-Duration: 1 Month

Advanced Technology Industrial Center (ATIC)

- ▶ Completed structured industrial attachment with exposure to modern machining and manufacturing processes
- ▶ Gained hands-on experience with industrial-grade equipment, production workflows, and quality control practices
- ▶ Observed CNC machining, metal fabrication, and material processing operations in a industry.

LEADERSHIP & ACTIVITIES

Coordinator — Creative & Media -*IMechE RUET Student Chapter*

- ▶ Led all digital content creation and creative media operations for the RUET student engineering chapter
- ▶ Developed professional engineering-focused content to strengthen chapter visibility and outreach
- ▶ Coordinated cross-team communication and managed event promotion campaigns

AREAS OF INTEREST

Industrial Automation & Smart Manufacturing | **Robotics & Autonomous Systems** | **Renewable Energy Systems** | **Additive Manufacturing (3D Printing)** | **Sustainable Bio-Composite Materials**

REFERENCES

Available upon request from thesis supervisor and faculty, Department of Mechanical Engineering, RUET.